# Session 2

## Problem Identification & Feasibility Analysis

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| **Course:** | Software Engineering Tools Lab (IT-B, 2026-27) |
| **Project:** | AI Interview Preparation Platform |
| **Team:** | Hirdesh (Lead), Mahek Bhatia, Janhavi Srivastava |
| **Deliverable:** | Problem Statement Document |
| **Industry Mapping:** | Product Requirement Analysis |

## 1. Session Objectives

This session focuses on the critical first phase of any software project: understanding the problem space and evaluating whether a proposed solution is viable. By the end of this session, students will be able to:

- Identify real-world software problems by analysing user pain points and market gaps.
- Write a clear, structured problem statement for a software engineering project.
- Conduct technical, operational, and economic feasibility analysis.
- Define the scope and boundaries of a software project.
- Map problem identification to industry product requirement analysis practices.

## 2. Identification of Real-World Software Problems

### 2.1 Why Problem Identification Matters

The success of any software product depends on how well the underlying problem is understood. A poorly defined problem leads to vague requirements, scope creep, wasted resources, and ultimately a product that fails to meet user needs. In industry, product managers and engineering teams invest significant effort in problem discovery before writing a single line of code.

### 2.2 Sources of Real-World Problems

Software problems can be identified from a variety of sources:

- **User Pain Points:** Direct feedback, surveys, interviews, and support tickets reveal frustrations users face with existing tools or processes.
- **Market Gap Analysis:** Studying competitors and the market landscape to find underserved needs or areas where existing solutions fall short.

- **Domain Expert Consultation:** Working with professionals in a specific field (healthcare, education, finance) to uncover workflow inefficiencies.
- **Technology Trends:** Emerging technologies (AI/ML, blockchain, IoT) create opportunities to solve problems that were previously infeasible.
- **Personal Experience:** Developers often build solutions for problems they have encountered themselves.

### 2.3 Problem Identification for Our Project

Through research and personal experience, our team identified the following problem in the interview preparation space:

- Job candidates often lack access to realistic, personalised mock interview practice.
- Existing platforms provide generic questions without real-time, AI-driven feedback.
- Resume screening by ATS systems is opaque - candidates do not know why they are rejected.
- There is no unified platform that combines mock interviews, resume analysis, and performance tracking in one place.

## 3. Problem Statement Writing

### 3.1 What is a Problem Statement?

A problem statement is a concise, formal description of the issue that the software project aims to solve. It serves as the foundation for all subsequent development activities - requirements, design, implementation, and testing all trace back to the problem statement.

### 3.2 Characteristics of a Good Problem Statement

- **Specific:** Clearly defines what the problem is, who it affects, and where it occurs.
- **Measurable:** Includes criteria that can be used to determine if the problem has been solved.
- **Relevant:** Addresses a genuine need for a defined set of users or stakeholders.
- **Scope-Bounded:** Does not try to solve everything; focuses on a tractable subset.
- **Solution-Neutral:** Describes the problem without prescribing a specific solution.

### 3.3 Problem Statement Template

A structured problem statement typically follows this format:

```
WHO:        Who is affected by the problem?
WHAT:       What is the problem they face?
WHERE:      In what context or environment does it occur?
WHEN:       When does the problem manifest?
WHY:        Why is it important to solve?
IMPACT:     What are the consequences of not solving it?
```

### 3.4 Our Problem Statement

> **Problem Statement: AI Interview Preparation Platform**
>
> Job seekers and career changers face significant challenges in preparing for technical and behavioral interviews. They lack access to personalised, AI-driven mock interview practice that simulates real interview conditions and provides actionable feedback. Existing solutions offer static question banks without real-time analysis of communication skills, technical accuracy, or response quality. Additionally, candidates struggle with ATS-optimised resume formatting, often resulting in qualified applicants being filtered out before reaching human reviewers. There is a need for a unified, intelligent platform that combines realistic interview simulation, instant performance analytics, and resume compatibility analysis to help candidates improve their interview readiness and increase their chances of securing employment.

## 4. Technical and Operational Feasibility

### 4.1 What is Feasibility Analysis?

Feasibility analysis evaluates whether a proposed project is practical and achievable given the available resources, technology, and constraints. It is a critical step before committing to full-scale development and helps stakeholders make informed go/no-go decisions.

### 4.2 Types of Feasibility

There are three primary dimensions of feasibility analysis:

#### A. Technical Feasibility

Assesses whether the required technology, tools, and technical expertise are available to build the proposed system.

| Factor | Assessment |
|---|---|
| Frontend Technology | Next.js + Tailwind CSS - mature, well-documented frameworks |
| Backend Technology | FastAPI (Python) - excellent for AI/ML integration |
| AI/ML APIs | Gemini API, OpenAI API - production-ready, well-supported |
| Database | PostgreSQL + Redis - industry-standard, scalable |
| Team Expertise | Team has skills in web dev, Python, and AI/ML |
| Development Tools | Git, GitHub, VS Code - readily available |

**Verdict: Technically Feasible**

#### B. Operational Feasibility

Evaluates whether the system can be effectively used and maintained in its intended environment.

- Target users (job seekers, students) are comfortable with web-based applications.
- The platform requires minimal training - intuitive UI with guided workflows.
- AI-generated feedback is easy to understand and actionable.
- The system can be deployed on cloud platforms (AWS/GCP/Vercel) for 24/7 availability.
- Maintenance involves updating question banks and fine-tuning AI models periodically.

**Verdict: Operationally Feasible**

### C. Economic Feasibility

Determines whether the project can be completed within the available budget and resources.

| Cost Component | Estimate |
|---|---|
| AI API Usage (Gemini/OpenAI) | Free tier / minimal cost for academic use |
| Cloud Hosting (Dev Phase) | Free tier (Vercel, Railway, or GCP) |
| Database Hosting | Free tier (Supabase / Neon PostgreSQL) |
| Development Tools | Open source / free for students |
| Domain Name (Optional) | ~$10-15/year |

**Verdict: Economically Feasible**

## 5. Project Scope Definition

### 5.1 What is Project Scope?

Project scope defines the boundaries of the project - what will be built (in-scope) and what will not be built (out-of-scope). A well-defined scope prevents scope creep, sets clear expectations for stakeholders, and helps the team focus on delivering a viable product.

### 5.2 In-Scope Features

1. AI-powered mock interview engine with technical and behavioral question generation.
2. Real-time feedback on response quality, communication pacing, and technical accuracy.
3. Resume upload and ATS-compatibility analysis with improvement suggestions.
4. User dashboard with performance history, scores, and progress tracking.
5. User authentication and profile management.
6. Multiple interview domains: software engineering, data science, product management.

### 5.3 Out-of-Scope (Future Enhancements)

- Live video interview recording and body-language analysis.

- Mobile application (iOS/Android) - web-first approach for this phase.
- Peer-to-peer mock interview matchmaking.
- Integration with job boards (LinkedIn, Indeed) for direct job applications.
- Multi-language support beyond English.

### 5.4 Constraints and Assumptions

Constraints:

- Development timeline is limited to one academic semester.
- Team size is fixed at three members.
- AI API usage is constrained by free-tier or academic-license limits.

Assumptions:

- Users have a stable internet connection and a modern web browser.
- AI APIs (Gemini, OpenAI) will remain available with current pricing/free-tier limits.
- The team will follow Agile sprints with bi-weekly deliverables.

## 6. Deliverable: Problem Statement Document

The Problem Statement Document for this session consolidates the analysis performed above into a formal deliverable. It includes:

1. Problem identification and background research on the interview preparation domain.
2. Formal problem statement following the WHO-WHAT-WHERE-WHEN-WHY-IMPACT structure.
3. Technical, operational, and economic feasibility assessment with verdicts.
4. Project scope with in-scope features, out-of-scope items, and constraints.

Document location in the repository:

```
2026-27_IT-B_202401100600089/
+-- session/
    +-- Session 2 - Problem Identification & Feasibility Analysis.pdf
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## 7. Industry Mapping: Product Requirement Analysis

The activities in this session map directly to the Product Requirement Analysis phase used in the software industry:

- **Product Discovery:** Companies like Google and Amazon run design sprints and user research to identify problems before building solutions.
- **PRD (Product Requirements Document):** Product managers write PRDs that begin with a

problem statement - the same skill practised here.

- **Feasibility Reviews:** Engineering teams conduct technical feasibility reviews before sprint planning to assess risk and resource requirements.
- **Scope Planning:** Agile teams define MVP (Minimum Viable Product) scope using techniques like MoSCoW prioritisation (Must, Should, Could, Won't).
- **Business Case Analysis:** Startups and enterprises alike evaluate economic feasibility through cost-benefit analysis, ROI projections, and resource planning.
- **Stakeholder Alignment:** Problem statements and scope documents are shared with stakeholders (clients, management, investors) to ensure alignment before development begins.

## 8. Session Summary

In this session, we identified the real-world problem of inadequate AI-driven interview preparation tools, crafted a formal problem statement, conducted a comprehensive feasibility analysis across technical, operational, and economic dimensions, and defined the project scope with clear in-scope and out-of-scope boundaries. This foundational work ensures that the subsequent design and development phases are grounded in a well-understood problem space.

Next Session: Requirements Engineering - SRS Document and Use Case Modelling.